ABNORMAL PATTERNS IN CONTINUOUS PATIENT MONITORING

Is it the patient or the sensor? Separating artifacts from real events in vital-sign streams

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PROBLEM STATEMENT

Bedside monitors raise many alarms that are not about the patient: a flushed arterial line or a moved probe creates dramatic readings that mean nothing. Real deterioration rarely shows in one signal alone.

Can we tell a sensor artifact from a real physiological change?

OBJECTIVES

1. Clean and explore real monitor data
2. Understand how the five vitals behave and relate
3. Flag abnormal episodes and name each one: artifact, real event or review
4. Test it against baselines and report its limits

KEY FEATURES

- Domain-aware cleaning, no median imputation
- Outliers checked against clinical ranges, not deleted blindly
- Patient-specific baseline: each patient is compared with their own last 5 minutes
- Multi-vital rule: signals that move together are real
- Held-out test with injected ground truth and two baselines

DATASET

VitalDB, Seoul National University Hospital (Lee et al., *Scientific Data* 2022). 30 random surgical cases, **199,876 readings** at 2-second resolution.

HR	SpO_2	MAP	Temp	RR
beats/min	%	mmHg	°C	breaths/min

TECHNOLOGIES

Python · pandas · NumPy · SciPy · scikit-learn · Matplotlib · Seaborn · vitaldb · Google Colab

FUTURE SCOPE

- Clinician-labelled artifacts for real evaluation
- Learned sequence models and ICU data (eICU)
- Artifact sources that hit several vitals at once, e.g. electrocautery
- Live alert dashboard

METHODOLOGY

1. **Load** 30 cases with all five vitals, 2 s grid
2. **Clean** impossible values and probe warm-up removed; short gaps bridged per patient
3. **Explore** missing values, IQR outliers vs clinical ranges, correlations, time series
4. **Detect** flag a reading when it leaves the patient's 5-minute baseline by more than K × a robust scale (K = 4)
5. **Classify** each episode by what the other vitals do within ±30 s
6. **Evaluate** inject known artifacts and events into held-out patients (70/30 split); compare with Isolation Forest and clinical thresholds

- **Artifact** isolated on one vital, ≤ 30 s
- **Real event** several vitals move together
- **Review** one vital, sustained: unclear

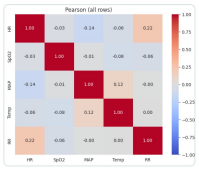
RESULTS: CLEANING AND EDA

70% of rows complete (139,501 of 199,876)

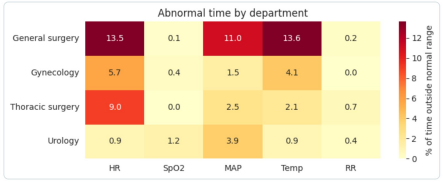
18,153 impossible readings removed, mostly MAP below 20 mmHg

25% of rows an IQR purge would have deleted; kept instead

1.1% of SpO_2 IQR outliers are clinically abnormal (HR, Temp, RR: 100%)

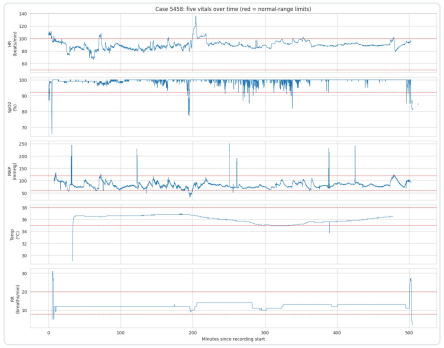


Vitals are only weakly correlated (strongest: **HR–RR, r = 0.22**). No single-vital rule can separate artifact from event.



Share of time outside the normal range, by department. General surgery is highest for HR, MAP and Temp.

RESULTS: ONE PATIENT, TWO KINDS OF EVENT



Isolated **MAP spikes above 200 mmHg** leave every other vital untouched: artifact. Near minute 200, **HR rises, SpO_2 falls and MAP dips together**: a real event.

RESULTS: DETECTOR EVALUATION

On the 30 real recordings (K = 4)

1,563 episodes in 111 recorded hours

53 · 36 · 11 % artifact · real event · review

51% of episodes are on HR (798)

Held-out test with injected ground truth

100% · 84% artifacts (69): found · named correctly

100% · 100% real events (31): found · named correctly

0.42 row-level F1. Isolation Forest 0.43, clinical thresholds 0.20

9 held-out patients, 34.8 h. Isolation Forest flags anomalies but cannot say artifact or event. 11 artifacts were mistaken for real events.

KEY INSIGHTS

- **MAP is the dirtiest signal:** 6.8% of its readings were physically impossible (arterial-line flushes).
- **Statistical outlier ≠ clinical event.** SpO_2 sits at 100%, so 18% of rows are "outliers" but only 1.1% are truly low.
- **Agreement between vitals** is the only reliable clue, because single-vital correlations are weak.
- **On the injected test every artifact and event was found** and 89 of 100 were named correctly. Isolation Forest has a similar F1 but cannot tell which is which.

CONCLUSION

Careful cleaning plus a patient-specific, multi-vital rule gives a transparent way to flag likely artifacts without discarding real clinical events. Limits: evaluation uses injected, not clinician-labelled, truth, and precision is low (0.27) because natural anomalies are unlabeled.